Physics 101: Mechanics

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Contents

Chapter 1

The Distance Formula

EXPERIMENTAL LAB: FIELD APPLICATIONS

1. **Laser Rangefinder Calibration:** A surveyor places a laser at origin $(0, 0)$ and a reflector at $(15, 20)$. The laser reads 24.8 units. Calculate the theoretical distance and find the percentage error.
2. **The Acoustic Localization:** Two microphones at $M_1(-5, 0)$ and $M_2(5, 0)$ detect sound. A third at $M_3(0, 12)$ detects it later. Find the source $S(0, y)$.
3. **Shadow Tracking:** A pole's top is at $(0, 5)$ and its shadow is at $(3, 0)$. When the sun moves, the shadow moves to $(0, 0)$. Calculate the total distance the shadow tip traveled.
4. **Tension Wire Stability:** A tower at $(0, 12)$ is secured by wires anchored at $(x, 0)$ and $(-x, 0)$. If the total length of both wires is 26, find x.
5. **GPS Drift Analysis:** Readings at $(10, 10)$ drift to $(10.1, 9.9)$ and $(9.8, 10.2)$. Find the average distance of these drift points from the center.

Chapter 2

Coordinate Geometry

2.1 Introduction

Coordinate geometry, also known as analytic geometry, is the study of geometry using a coordinate system. This contrasts with synthetic geometry.

2.2 The Cartesian Plane

The Cartesian plane is defined by two perpendicular number lines: the x-axis, which is horizontal, and the y-axis, which is vertical.

Chapter 3

New Chapter

EXPERIMENTAL LAB: ANNIHILATOR OPERATOR

Definition: An *annihilator* is a differential operator that maps a function completely to zero when applied to it.

Let

$$D = \frac{d}{dx}.$$

1. **Differentiate the Function:** For the exponential function e^{2x},

   $$D(e^{2x}) = \frac{d}{dx}(e^{2x}) = 2e^{2x}.$$

   Hence, the derivative is still a multiple of the original function, so D alone does not annihilate it.

2. **Apply the Annihilator:** Consider the operator $(D - 2)$:

   $$(D - 2)e^{2x} = D(e^{2x}) - 2e^{2x} = 2e^{2x} - 2e^{2x} = 0.$$

   Therefore, the annihilator of e^{2x} is

   $$\boxed{D - 2.}$$

3. **Meme Interpretation:**

   - **Using $D = \frac{d}{dx}$:** Shoots e^{2x}, but it clones itself since

     $$D(e^{2x}) = 2e^{2x}.$$

   - **Using $(D - 2)$:** Actually annihilates e^{2x} because

     $$(D - 2)e^{2x} = 0.$$